CHAPTER 1

INTRODUCTION

Road transportation has become an important component of modern life facilitating economic growth, mobility, and social interaction across the globe However, alongside these benefits lies the persistent challenge of road traffic incidents, which continue to cause significant loss of life, property damage, and social disruption in both developed and developing nations. Despite major advancements in vehicle technology, infrastructure development, and road safety regulations, traffic-related accidents remain among the most critical public health and safety concerns worldwide. According to the World Health Organization (2023), an estimated 1.19 million people die each year due to road traffic accidents, making it the leading cause of death among individuals aged 5 to 29 years old. A large proportion of these fatalities involve vulnerable road users such as motorcyclists, cyclists, and pedestrians underscoring global gaps in road safety infrastructure and enforcement.

In the Philippine context, these global challenges are reflected in the country's ongoing struggle to establish efficient, real-time systems for traffic incident reporting and management. Current mechanisms remain inadequate to

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meet the demands of accurate information sharing and rapid emergency response Although various mobile applications and digital tools have been introduced to enhance reporting efficiency, many of these technologies are not widely adopted or consistently utilized. As a result, most incidents are still documented manually heavily relying on the individual's ability to describe events and locations accurately. This dependence on human input often leads to reporting delays and inaccuracies, which hinder the timely delivery of emergency services and the formulation of responsive traffic management strategies.

The limitations of manual reporting systems highlight the pressing need for more reliable, integrated, and accessible mechanisms across the country. Incident reporting in the Philippines faces multiple barriers, including the fear of reprisal among witnesses and victims, lack of community cooperation, technological constraints such as poor internet connectivity, and even the suppression of reports through libel laws or cyberattacks. Additionally, insufficient local government capacity, fragmented resources, and threats of violence from both state and non-state actors further weaken the effectiveness of reporting and data management.

Recent studies reinforce these concerns by emphasizing the inefficiencies of paper-based traffic incident documentation. The Philippine Statistics Authority (2023) reported over 34,000 road crashes nationwide in 2022. However, many police units still rely on outdated, manual reporting systems. These handwritten

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reports are frequently delayed, incomplete, and inconsistent, resulting in substantial data gaps that impede evidence-based policymaking. Similarly, the Department of Transportation (2022) noted that traditional crash report forms often omit vital information such as geographic0020coordinates, weather conditions, and exact times of occurrence, making comprehensive analysis difficult. The Global Status Report on Road Safety further emphasizes that countries dependent on manual reporting systems are prone to data loss, transcription errors, and inconsistencies, undermining road safety enforcement and prevention efforts.

These national-level deficiencies are likewise evident in local settings such as Lipa City, where traffic incident management remains largely manual. Paper forms, phone calls, and fragmented record-keeping continue to be the primary modes of documentation among police, traffic enforcers, health responders, and towing services. This outdated system often produces incomplete or delayed reports, leading to slower response times, weakened evidence preservation, and missed opportunities for data-driven road safety interventions. Moreover, the absence of integrated digital infrastructure restricts the city's ability to collect, analyze, and share critical traffic data across agencies. Without standardized and accessible digital records, it becomes difficult to identify accident hotspots, monitor repeat offenders, or evaluate the effectiveness of current road safety measures.

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The lack of interoperability and coordination among multiple stakeholders including police departments, emergency medical services, insurance companies and city engineering offices further exacerbates inefficiency. Each agency tends to maintain separate records, resulting in data duplication, miscommunication, and difficulties in long-term planning or legal verification. Consequently, the fragmented system hinders comprehensive analysis of traffic trends and prevents the development of coordinated response strategies that could significantly enhance road safety outcomes

To address these issues, the proposed implementation of TRACKFORCE LIPA aims to revolutionize traffic incident reporting and management through the integration of mobile-based technologies. This digital platform seeks to enhance communication between field personnel and command centers by providing real-time location tracking, automated data entry, and centralized reporting. Through its ability to generate detailed reports and maintain audit trails, TRACKFORCE LIPA promotes accountability, transparency, and evidence-based decision-making. The platform's data-driven insights can further support academic research, urban planning initiatives, and policy development, potentially serving as a model for other municipalities facing similar challenges.

This study is therefore conducted to identify and analyze the inefficiencies and gaps within the current manual traffic incident reporting and investigation

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systems of Lipa City. It aims to establish a baseline understanding of existing processes, providing empirical evidence on how these limitations impact operational effectiveness and public safety. Furthermore, the study will contribute to the development of a digital solution that improves coordination, efficiency, and data reliability. By generating valuable information for policy formulation and traffic management strategies, this research also contributes to the broader discourse on digital transformation in municipal services. The findings of this study will emphasize how the integration of technology through systems like TRACKFORCE LIPA can address underreporting, reduce response time, and strengthen public trust in local authorities, thereby promoting safer and more efficient roads for all.

Research shows that implementing digital platform can successfully reduce traffic incident reporting and accident investigations while also enhancing accountability and fostering trust in institutions. These efforts align with SDG 16, which supports the development of peaceful, inclusive societies, access to justice for everyone, and the building of efficient, accountable, and transparent institutions. Thus, digital systems serve as tools that legitimize institutions by making government activities transparent and verifiable.

On the other hand, this digital platform aims to improve communication between field staff and command centers through real-time location tracking, automated data input, and centralized reporting. Using digital tools connects the

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goals of SDG 9, which focuses on resilient innovation, with those of SDG 16, emphasizing peace and accountability. This demonstrates how technology can directly support communities in becoming both resilient and sustainable.

The findings of this research will not only explain the introduction of a mobile based incident reporting system in Lipa City but also contribute to broader discussions on how technology can be harnessed to address under-reporting, improve response time, and enhance trust between communities and authorities. This linkage between technological innovation and public safety provides a strong rationale for the conduct of this study

CONCEPTUAL FRAMEWORK

The present study on TrackForce Lipa**:** A Digital System for Traffic Incident Reporting and Accident Investigation is anchored on theories and prior studies that emphasize the role of technology in enhancing reporting systems, improving data accuracy, and strengthening institutional response.

According to Ignaco (2021) in the Journal of Informatics and Virtual Education (JOIV), the use of mobile applications in incident reporting allows users to submit accurate details such as GPSbased location, incident description, and supporting images. This innovation addresses the common challenge of

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incomplete or inaccurate information in manual reporting. Their study revealed that digital incident reporting systems in the Philippines received positive feedback, particularly in terms of user acceptance and usefulness.

Similarly, Bulusan (2019) in the International Scholars Conference Proceedings developed a mobile application for first responders that integrated GPS tracking, image capture, and automatic location transmission. The system was evaluated highly in terms of accuracy, error reduction, and ease of use, highlighting the importance of user-friendly interfaces and real-time functionalities in incident management systems. Both studies employed user feedback as an essential component in refining the system, showing the significance of continuous evaluation and prototyping in software development.

The theoretical underpinning of this study is also guided by the Technology Acceptance Model (TAM) (Davis, 1989), which posits that Perceived Usefulness and Perceived Ease of Use are critical determinants of user acceptance of new technologies. In the context of TrackForce Lipa, the usefulness of the system is measured by its ability to improve reporting efficiency, enhance data accuracy, and ensure timely responses, while ease of use pertains to its interface design and accessibility to both officers and citizens. Furthermore, the framework adopts the System Quality Model (ISO/IEC 25010), which emphasizes core attributes such as functionality, reliability, usability, and efficiency. These attributes directly influence

user satisfaction and system adoption. In this study, system quality is manifested in the accuracy of GPS-based reporting, reliability of the digital police blotter, and the efficiency of generating and transmitting reports.

Finally, the study is informed by the Barrier–Facilitator Theory of Incident Reporting (Archer et al., 2017), which identifies factors that hinder or encourage individuals to engage in reporting systems. Barriers may include fear of reprisal, lack of knowledge, or technical difficulties, while facilitators involve management support, training, and system simplicity. These aspects are relevant in the Philippine law enforcement setting, where both institutional and technological readiness shape the adoption of digital reporting systems.

Based on these theoretical and empirical foundations, the framework assumes that:

1. System Quality (accuracy, usability, reliability) influences

2. User Perspective (usefulness, ease of use, trust), which in turn affect

3.1 System Adoption and Utilization, leading to

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3.2 Enhanced Outcomes (timely reporting, improved accuracy, effective law enforcement response)

INPUT	PROCESS	OUTPUT
Current Condition of manual system in term of accuracy, usability, reliability Problems encountered with the use of manual system Root causes of identified problems in the use of manual system Perceived impact of the proposed digital system as an alternative	Interview Survey Questionnaire Data analysis	Digital System as an Efficient Alternative to the Manual System TracKForce Lipa

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STATEMENT OF THE PROBLEM

This study aims to quantitatively assess the effectiveness of the manual reporting system in traffic incident documentation and accident investigation and to evaluate the acceptability of the proposed digital reporting system, TrackForce Lipa.

Specifically, this study seeks to answer the following questions:

1. What is the demographic profile of the respondents in terms of:

1.1. Age

1.2. Sex

1.3. Type of Vehicle Involved

1.4. Number of Times Involved in Traffic Accident in Lipa City

2. To what extent does the manual reporting system affect processing time during traffic incident documentation and accident investigation, as measured in terms of:

2.1. Incident information collection time

2.2. Evidence documentation time

2.3. Report completion and submission time

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3. What is the level of accuracy and completeness of evidence documentation under the manual system, in terms of:

3.1. Frequency of documentation errors

3.2. Incidence of missing or incomplete information

3.3. Retrieval success rate of case records

4. To what extent does the manual system affect the integrity and preservation of evidence, as measured by the frequency of:

4.1. Lost data

4.2. Altered or tampered data

4.3. Delayed data transfer to legal or administrative units

5. What is the level of acceptability of TrackForce Lipa as a digital reporting system, as perceived by traffic investigators, measured in terms of:

5.1. Perceived usefulness

5.2. Perceived ease of use

5.3. System reliability

5.4. User intention to adopt

6. What challenges are encountered by traffic investigators under the manual reporting system, quantified in terms of:

6.1. Incidence of inconsistent report formats

6.2. Percentage of reports with delayed submission

6.3. Data retrieval and storage performance

HYPOTHESIS

H_0: The TrackForce Lipa digital system has no significant effect on the efficiency, accuracy, or timeliness of traffic incident reporting and accident investigation in Lipa. Alternative Hypothesis (H_1)

H_1: The TrackForce Lipa digital system significantly improves the efficiency, accuracy, and timeliness of traffic incident reporting and accident investigation in Lipa.

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SCOPE, LIMITATION AND DELIMITATION

This study focuses on the design, development, and evaluation of a Digital System for Incident and Accident Investigation intended for use by the Philippine National Police (PNP) Lipa. This investigation will be conducted exclusively within the territorial jurisdiction of Lipa City, Batangas. All data collection activities such as surveys and interviews will take place only in this locality. Limiting the geographic area ensures that the findings accurately reflect the conditions and operational environment of traffic incident reporting in Lipa City and are not generalized to other municipalities or provinces

This research focuses solely on Philippine National Police (PNP) traffic investigators assigned within Lipa City, Batangas. This Study is limited to the development and implementation of the system within the PNP Lipa jurisdiction Only personnel who directly handle traffic incident reporting and accident investigation are included as respondents. Other PNP units such as patrol, intelligence, or administrative divisions are deliberately excluded from the study Likewise, motorists, commuters, and community members outside Lipa City are not part of the participant group

This research will conduct the data through interview and questionnaires with the use of Google Forms (G-FORMS). This clear delimitation ensures that the

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findings represent the perceptions of those most directly involved in the intended use of the Track Force Lipa system. The system's coverage is confined to the documentation and management of data directly relevant to vehicular accidents, traffic violations, and related incidents. These delimitations ensure that the research remains feasible within the given timeframe and available resources while maintaining a clear focus on enhancing digital incident and accident investigation processes.

SIGNIFICANCE OF THE STUDY

This study entitled " TrackForce Lipa: A Digital System for Traffic Incident Reporting and Accident Investigation "is significant to following:

To **Lipa City Police Station**, this study is significant to the PNP Lipa as it provides and innovative digital system that can improve the accuracy, efficiency, and timeliness of incident and accident investigation. It can help PNP LIPA By shifting from a manual to a digital platform, the system is envisioned to reduce delays, minimize errors, and provide a more standardized process in documenting cases. As a result, the PNP Lipa can respond faster to emergencies and ensure that investigation is based on reliable and updated data

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To the **Road Users**, this study can help to enhance traffic incident report and accident investigation by providing a centralized platform where traffic incidents can be reported in real time, investigated systematically, and stored securely for future reference. The digital system offers a safer and more transparent traffic environment. This study provides a faster reporting and documentation of road incident. The road user can benefit from timely updates and improved traffic management.

To the **Future Researchers and Lipa City Police Station Traffic Investigation Unit**, this study can serve as an extra source of ideas about TrackForce Lipa: A Digital System for Traffic Incident Reporting and Accident Investigation. It may provide them with information, a guide for their own studies, and a reference if they plan to do similar research in the future. This can also help them improve and widen their reports

In wider context, this research supports the progress of digital transformation within public service. By integrating technology into police operations, it sets an example how digital system can be used to improve governance, safety, and accountability. The effective deployment of this system in PNP Lipa can serve as a benchmark for other law enforcement units nationwide, promoting a more intelligent and data-driven strategy for public safety and traffic management.

DEFINITION OF TERMS

The terms considered essential for understanding the research were included in the manuscript and thus were defined both operationally and conceptually. This means the terms were clarified by explaining their theoretical meaning as well as how they are specifically measured or applied within the study.

Administrative Division. This refers to a specific area or unit of government that is managed by local authorities for administrative and organizational purposes. A geographical area within a country that is established for government management purposes, distribute governmental responsibilities across regions, making public services more efficient and accessible.

Barrier facilitator theory. a framework or concept used across various fields to identify factors that prevent or help a desired behavior, implementation, or outcome. A framework or concept used across various fields to identify factors that prevent or help a desired behavior, implementation, or outcome.

Geographic. This term refers to anything related to the physical area, location, or boundaries of a place. In this study, it describes the specific location where the research is done, which is Lipa City, Batangas. Refers to anything

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related to the physical area, location, or boundaries of a place. the specific physical region or location where a business, project, or military mission conducts its activities

Geographic Dispersion. the state or act of being spread out or scattered over a wide area, rather than being concentrated in one place. Extent to which a company's facilities, assets, operations, or resources are spread out across different geographic locations.

Pervasive risk. is a widespread threat that affects many areas. It can cause significant harm if not managed properly. A type of risk that is widespread and inherent across an entire organization, arising from ineffective or failed internal processes, people, systems, or external events that disrupt the core business operations.

Roadway. The full width of a public way or place used for vehicular traffic. A roadway can be defined by its structural and functional characteristics that support continuous and controlled movement of vehicles along fixed routes, facilitating mobility and logistics in both urban and rural settings.

Road transportation. Is the movement of people or goods using vehicles on roads, offering unmatched flexibility, accessibility, and door-to-door service for

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short to medium distances, but facing challenges with high energy consumption and emissions. Refers to the actual movement of vehicles involved in traffic incidents and accidents within the study area, as recorded and processed through the digital incident and accident reporting system.

Territorial Jurisdiction. Refers to a geographic area or physical boundaries within a particular authority or institution has the legal power to enforce laws, investigate incidents and exercise duty. Specific geographic area of responsibility where traffic incidents and accidents are officially investigated, documented, and managed by authorized law enforcement personnel using the system

Traffic Incident Reporting. This refers to the process of recording and documenting any road-related event such as accidents, collisions, or other traffic disturbances. In this study, it means the collection of details like time, location, people involved and causes by the Philippine National Police (PNP) traffic investigators.

Traffic Management. The process of overseeing, coordinating, and controlling traffic, including pedestrians and vehicles, to ensure safety and efficiency. It includes measures such as traffic signal control, incident management, flow optimization, regulation enforcement, and use of technologies

like intelligent traffic systems to maintain road capacity, safety, and reliability within the transport network.

Traffic Way. A broader term for any public path or route used for travel. A premises or area open to the public by right or custom for the use of motor vehicles.

CHAPTER 2

REVIEW OF RELATED LITERATURE

This literature review provides background and contextual information to better understand materials relevant to the present study on digital traffic incident reporting systems.

Related Literature

Incident Management System

(Abuan, K.F.A. *et al.* 2023), This study presents the automation of the manual process of collecting and analyzing incident data and transparency with the general public. The output of the study will be a web application system that will produce descriptive analytics, hot spots, and forecasting (7MA), supporting the PNP's preparation for the projected future incident cases in the following week and determining the hazardous area in Pasig City through the hot spots. The general public will be informed of the number of road incidents in a particular barangay, entire cases in Pasig City, and other data visualizations integrated into the dashboard. Scrum Methodology was used to construct the system, and numerous users with varying responsibilities and administrator permissions were necessary to access and control the system.

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Traffic violation accident information reporting system

According to (Zhang et al., 2020) article, a traffic violation accident information reporting system and method based on an automobile data recorder is described, where the system comprises a storage module and a video information collection module of the car data recorder the system further comprises a wakeup module, an information generation module, video information acquisition module, vehicle positioning module, information integration module, and a communication module. The problems that the public supervision is complex in reporting and the traffic violation accident supervision is incomplete in the prior art are solved.

Android Based Real Time Road Accident Reporting Application

(Villanueva, 2022), this study is conceptualized to develop an android-based real-time road accident reporting application using your android mobile devices. It is a mobile application that captures the accident and automatically gets the coordinates or the location of the accident and sends it to the nearest PNP office. In addition, the application requires the user to input their full name and contact details for validation. Furthermore, the conforming office will receive a notification on the web application and validate the report. With the current increase in the development of transportation, road widening projects, and motorcycle loops, the community has been suffering from road-related accidents, especially in an unfamiliar and isolated area.

Mobile Application for Incident Reporting

According to (Ignaco 2021), In the Philippines, reporting an incident always depends on self-reporting to the nearest law enforcer's office or calling a channel using a mobile phone. 911 is the National Emergency hotline to get assistance when an emergency occurs. However, the emergency hotline operated by the Emergency Network Philippines (ENP), cannot retrieve the reporter's location details immediately. Only when the reporters describe the exact location clearly Yet, many circumstances that the reporters do not know when they are, or sometimes they have imprecise position information.

Road Incident Detection Leveraging Live Traffic Data

(Gannina 2024), The study in Procedia Computer Science (2024) introduces a digital system that detects and reports road accidents by analyzing real-time traffic data through APIs and GPS-enabled devices. By continuously monitoring traffic flow and identifying irregular patterns such as road closures or sudden stops, the system can instantly alert users and authorities, improving response times and road safety. It also helps law enforcement pinpoint accident-prone areas and high-risk drivers, enabling targeted safety interventions. This can help law enforcement agencies to identify drivers at risk of causing accidents and implement education and enforcement programs targeted towards them. Hence, the proposed approach can aid in reducing the negative impact of traffic.

Novel Internet of Things-Enabled Accident Detection and Reporting System

(Bhatti et al., 2020), The proposed approach aims to take advantage of advanced specifications of smartphones to design and develop a low-cost solution for enhanced transportation systems that is deployable in legacy vehicles. In this context, a customized Android application is developed to gather information regarding speed, gravitational force, pressure, sound, and location. The speed is a factor that is used to help improve the identification of accidents. The information acquired is further processed to detect road incidents. The proposed approach is validated through simulations and comparison with a real data set of road accidents acquired from Road Safety Open Repository, and shows promising results in terms of accuracy.

Application for Contactless Traffic Violation

(Jose, J. A. C., 2021), presents a prototype of a centralized contactless traffic violation apprehension system composed of an artificial intelligence (AI) engine and a web application. The AI engine collects traffic data, primarily traffic violation data, through a contactless approach by using different video and image processing techniques and AI algorithms in its three modules: license plate detection, optical character recognition (OCR), and number coding violation detection. The web application consolidates all the data produced by the AI engine and provides a graphical user interface (GUI) for data management, visualization, and analysis. This contactless apprehension system aims to automate,

standardize, and streamline the existing processes of law enforcement agencies and institutions for a more efficient apprehension of traffic violators and help them improve their traffic planning and management in the congested areas of the Philippines.

Traffic Management and Monitoring System

(Mariñas 2024), In this study, our focus was on creating an image recognition-based traffic and incident monitoring system utilizing video surveillance cameras for implementation in Urdaneta City. The following conclusions have been derived. The project requirements were meticulously analyzed by examining the existing business rules and policies of POSO Urdaneta City in incident monitoring implementation, influencing the design and development of the proposed system. Interviews with the Public Order and Safety Office in Urdaneta City emphasized the challenges in manual monitoring and the importance of adhering to safety rules. Collaboration with the PNP Urdaneta highlighted the need for timely responses to incidents. The study underscores the role of technology, collaboration, and efficient reporting in enhancing traffic management and public safety.

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Factors and Challenges in Road Incidents

(Segun, William, et. Al 2024), This study examined the factors that led to road incidents in Metro Manila, focusing on human behavior, public awareness and attitudes, road infrastructure and Mdesign, technological interventions, and the implementation of enforcement measures. The findings indicated that human error was the primary cause of road crashes, with speeding, driving under the influence of alcohol or drugs, distracted driving, failure to adhere to traffic rules, and aggressive driving behaviors identified as significant factors.

Data Analysis for Road Traffic Incident Response System

(Arispe, 2020). this study presents a use case diagram and system features to determine the accident-prone areas of the locality and to enhance the capacity of response units to respond to traffic incidents. The features of the study were designed to improve the notification rate of an incident, enabling the responding units to easily access the site and respond immediately. Integrating spatial data analysis was supported with the assessment of the accident-prone areas and reporting process on the mobile application of the emergency response system for traffic incidents. In order to develop a mobile application for traffic incident emergency response system and integrate spatial data analysis, this study present a use case diagram and system features to determine the accident-prone areas of the locality and to enhance the capacity of response units to respond to traffic incidents.

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Android Based Real Time Road Accident

Villanueva, C., & Palaoag, T. (2022). This study is conceptualized to develop an android-based real-time road accident reporting application using your android mobile devices. It is a mobile application that captures the accident and automatically gets the coordinates or the location of the accident and sends it to the nearest PNP office. In addition, the application requires the user to input their full name and contact details for validation. Furthermore, the conforming office will

receive a notification on the web application and validate the report. Afterward, the medical team will be notified for immediate assistance.

Data for Road Incident Visualization

(Burlacu, 2021). Robust crash data systems are key to understand the nature and magnitude of the road safety problem and to propose appropriate solutions. However, resource constrained governments oftentimes lack adequate data systems or tools for collecting, storing, managing, analyzing, and reporting historic crash data. As a result of previous World Bank efforts, and with the aim of tackling this problem, a free open-source system for recording and visualizing geo-referenced crash data was developed and piloted. This system, Data for Road Incident Visualization, Evaluation, and Reporting (DRIVER), does not have licensing costs for clients and can be deployed by agencies interested in improving their crash data management systems.

Effectiveness of Digital Innovation

(Escalona, 2023). This study was undertaken to focus on the effectiveness of digital innovation as the key to advanced road safety and investigation Specifically, aimed to determine the equipment being used by the PNP Highway Patrol Group, Bus Drivers and UV Express, including the effects of the equipment that are being used and the problems encountered in using the equipment that are being used.

Data Analysis for Road Traffic Incident

Arispe, M. C. A. (2020). This study presents a use case diagram and system features to determine the accident-prone areas of the locality and to enhance the capacity of response units to respond to traffic incidents. The result shows that it can be accessed on any mobile device in a user-friendly interface, accessible and useful to any community members for information about optimized routes to nearby response units. The user could easily report the accident and obtain the alert, while the response unit would review the report and determine the accident area coordinates. The features of the study were designed to improve the notification rate of an incident, enabling the responding units to easily access the site and respond immediately.

Related Study

Bautista et al. (2023) created “BALANGAY,” a web-based system streamlining document requests and incident reports within barangays, including decision support features through heatmap generation. This is anchored with the objectives of the study which is mainly to build a centralized database for barangays and design modules with the purpose of streamlining barangay processes. Part of it is to generate an incident heatmap that will assist barangay officials in creating better programs and project developments. Baltazar et al.

(2022) developed similar innovations for Hagonoy, Bulacan, integrating disaster and crime reports with decision support tools for local government units.

Rayno & Daily (2023) discuss how traffic crash investigators collect digital evidence from vehicle event data recorders. In severe crashes destroying typical network communications, more invasive techniques through in-circuit programming port or direct memory chip reading become necessary. Some of the data collected in a crash investigation is in the form of digital data from event data recorders built into the electronic control units in the vehicles. Occasionally, traffic crashes are severe enough to destroy the typical network-based communications protocols to extract the digital forensic data. In these cases, more invasive techniques of gathering forensic data through in-circuit programming ports or direct reading of data bearing memory chips are needed

Lontoc et al. (2023) evaluated PNP's electronic project systems, revealing that while digital tools improved record- keeping and transparency, they remain dependent on internet access and centralized servers, creating difficulties in rural areas. Many police stations continue relying on manual, paper-based methods prone to data loss, duplication, and delays. The study aimed to evaluate the implementation, and problems encountered in the implementation of Philippine National Police Electronic Project Systems in Lipa City Batangas. This study asserts that although the PNP's e-Project Systems in Lipa City Batangas are

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implemented, the regular conduct of capability building is necessary for the monitoring, supervision and management of the project.

According to PedSafe Philippines (2024), the Metropolitan Manila Development Authority (MMDA) launched comprehensive road safety action plans targeting 35% reduction in road crash fatalities by 2028. This road safety action plan is part of Project 6: Traffic Safety and Education under the 5-Year Action Plan of the Comprehensive Traffic Management Plan for Metro Manila, funded by the Japan International Cooperation Agency. The MMRSAP aims to enhance road safety and contribute to a more sustainable and livable environment in Metro Manila. The plan integrates data from the Metro Manila Accident Reporting and Analysis System (MMARAS) and the Data for Road Incident Visualization, Evaluation, and Reporting (DRIVERS) system.

Mercado M.T (2024) conducted qualitative research on traffic accident causes in Sta. Teresita Municipality, involving community members, Municipal Engineering office personnel, and Philippine National Police. The study found that traffic accidents are primarily influenced by human behavior factors, requiring comprehensive interventions for road safety education. It was found out that despite the interventions being conducted it needs a comprehensive way to assure that all the road users will be given enough time and attention to learn on road

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safety measures. The respondents also believed that through the intervention it helped them to be defensive drivers and become responsible to follow traffic rules.

Gomez et al. (2025) investigated technological resources and digital forensic capabilities of selected Anti- Cybercrime Units within the PNP. Their study revealed gaps in training, hardware availability, and technical expertise affecting investigation quality and speed. This study evaluates the technological resources and expertise within the Philippine National Police (PNP) Anti-Cybercrime Unit and their impact on cybercrime investigations. Findings reveal a significant positive correlation between the level of technological resources and the effectiveness of cybercrime investigations. Particularly, machine learning and digital forensics resources demonstrated strong correlations with both case resolution time and success rate. Additionally, proficiency, utilization, and adoption of technology by personnel showed significant positive impacts on investigation outcomes.

Ignaco (2022) developed mobile applications allowing users to report ordinance violations and public disturbances, automatically capturing GPS location data and supporting multimedia uploads. The barangay responder officer able to request a backup officer, the rescue emergency unit such as a hospital ambulance or firefighters, or transfer a report to the nearest police station. The system also manages web admin for responder locations and generates statistical reports including charts and graphs. A The positive feedback of the participants during

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the evaluation stage signifies that the application was accepted as tested and verified by the evaluation results.

Villarica et al. (2022) designed online crime reporting systems for Laguna incorporating analytics to identify crime patterns and hotspots for strategic resource allocation. The website offers analytics for decision-making support in addition to covering crime management information systems across police agencies for decentralization. Continued experimentation on index and non-index crime datasets to develop intelligent systems capable of forecasting must be explored, as the index crime dataset demonstrated a highly good result in the project. The system provides public access through geo-mapping that supports the community, or locally based crime prevention, instead of targeting individuals, targets areas where the risks of becoming involved in crime or being victimized are high.

Respocio & Co (2024) clarify that Philippine traffic accidents should ideally be reported to police immediately after occurrence. While no strict deadline exists for reporting, immediate reporting ensures accuracy and timely investigation initiation, maintaining fair representation for all parties and upholding public safety standards. The goal of traffic accident reporting is to ensure a fair and accurate representation of events for all parties involved, and to uphold public safety

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standards. Individuals are encouraged to treat the reporting process with the seriousness it warrants, irrespective of the timing.

The Philippine National Police announced a fully centralized digitalized crime-reporting app enabling Metro Manila residents to submit incident reports via mobile devices, expected to improve response times and overall efficiency. The DRIVERS platform, developed by DOST-ASTI, integrates information from local government units, hospitals, and law enforcement agencies, helping authorities identify crash-prone areas and respond more effectively to road incidents.

(Mercado 2024) study found out that the causes of traffic accidents are influenced by many factors generally were caused by human behavior that all the causes identified have something to do on how a road user reacts or acts in the road situations. Mercado MA.T. (2024) the study aimed to determine the Causes of Traffic Accidents in the Municipality of Sta. Teresita.

According to Villarica, Mai V. et al. (2023). online crime reporting system provides various functions and features for various users. This can be used to raise people's awareness regarding dangerous locations and help agencies predict future crime in a specific location within a particular time.

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Bulusan et al. (2020) designed a mobile application for incident reporting in Silang, Cavite. Allowing first responder to report incident through GPS tagging and photo documentation. The system addressed limitations of manual reporting methods and demonstrated faster response times and improves accuracy.

Godilo Kiseo (2021) Assessed the Philippine National Police's Crime Incident Reporting System (CIRS) and identified challenges such as limited connectivity and data delays. The Study emphasized the need for real-time reporting and integrated investigation tools, insights that are relevant to the present system design.

According to Burlacu, R. et al. (2022), the Data for Road Incident Visualization, Evaluation, and Reporting (DRIVER) system helps governments collect, manage, and visualize road-incident data. It was piloted in Cebu, Philippines, to improve road-safety planning through accurate, geo-referenced crash information.

According to Soytong, M. et al. (2025), a digital system using a data-matching algorithm can consolidate traffic incident data from multiple agencies in the Philippines, improving data accuracy and supporting evidence-based road safety interventions. The research focuses on creating a data validation and

matching algorithm to improve the integrity of this information, especially as agencies work towards creating a unified database for all road safety data.

(Kung et al., 2024) study introduces the Collision Clarification Generator (CCG), a Large Language Model-based system that streamlines traffic accident documentation through questioning, information extraction, and sequence generation, achieving high accuracy and completeness in human and LLM-based evaluations. The system employs predefined question templates and a standardized Traffic Accident Record Format (TARF) to ensure comprehensive data collection.

(Ahmed et al., 2024) paper presents a multimodal AI framework for real-time traffic incident response, integrating computer vision and natural language processing to detect accidents, coordinate emergency responses, and manage traffic flow, validated through diverse accident videos and simulation testing. These reports are automatically sent to relevant authorities, ensuring prompt and effective response. The system's effectiveness is validated through the analysis of diverse accident videos and zero-shot simulation testing within the environment.

(Huang, 2022) a spatial mobile crowdsourcing framework is proposed, enabling road users to report traffic events such as accidents, traffic rule violation,

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bad road and environment conditions through their mobile devices using images, videos and mobile sensor data. Timely detection of traffic events is of upmost importance in contributing towards traffic safety and ease of commute.

(Sneha et al., 2024) This review proposes a traffic congestion management and accident detection system to reduce congestion at junctions and to provide emergency assistance during accidents. The proposed system employs advanced computer vision and image processing techniques like You Only Look Once (YOLO) to monitor and analyze real-time traffic conditions and accidents. The pivotal feature of this system lies in its adaptive decision-making capability, automatically adjusting traffic signal timings based on observed density patterns and updating and reporting about the congestions and accidents for which Internet of Things (IOT) technologies are used.

(Jlil et al., 2024) study blockchain-based mobile application for traffic accident report management streamlines the process of reporting accidents, ensuring data authenticity, integrity, and reliability. An effective management of accident information is a crucial element, especially in the absence of an information system for amicable accident report management. This paper provides an innovative solution. It's a use case study of developing a Blockchain and interplanetary file system (IPFS) protocol-based system for traffic report

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management, using a mobile application. This application could be used at the time of the traffic accident.

According to Alincastre and Dalugdog (2022) evaluated the Philippine National Police's electronic project systems from the perspective of internal stakeholders. They found that digital platforms improve operational efficiency, data accuracy, and interdepartmental coordination while reducing reliance on manual records. Challenges such as limited user training and occasional system downtime were also noted, highlighting areas for improvement in digital adoption

(Villanueva & Palaoag, 2022). This study is conceptualized to develop an android-based real-time road accident reporting application using your android mobile devices. It is a mobile application that captures the accident and automatically gets the coordinates or the location of the accident and sends it to the nearest PNP office. It developed an android-based real-time road accident reporting application using your android mobile devices

Ignaco (2023) developed a mobile application for incident reporting in response to delays and inaccuracies in traditional manual reporting systems in the Philippines. The app allows users to submit incident details, photos, and GPS locations directly to authorities through a centralized digital platform, improving response time and data accuracy. The study demonstrated that mobile-based

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reporting enhances real-time communication, reduces reliance on paper-based methods, and ensures the reliability of submitted reports.

(Yoon et al., 2021) Paper attempts to propose a traffic investigation digital framework using digital data generated by devices and sensors equipped with self-driving cars and climate and rapidly changing road traffic information. Traffic accident investigation has been used to reconstruct the cause and condition of the accident by using images and driving record data that introduce IT technology, and to analyze both sides of the accident, skid marks, and vehicle damage to identify the perpetrators and victims.

(Pádua et al., 2020) methodology that aims to support road traffic authorities/professionals in activities involving the collection of data/evidences of motor vehicle collision scenarios by exploring the potential of using low-cost, small-sized and light-weight unmanned aerial vehicles (UAV). The reconstitution of road traffic accidents scenes is a contemporary and important issue, addressed both by private and public entities in different countries around the world.

(Pedraza-Farías et al., 2025) study presents a real-time traffic incident monitoring system for urban traffic management, leveraging Waze data and a spiral methodology to analyze and visualize traffic incidents, facilitating evidence-based decision-making and optimizing mobility in cities like Bogotá. This article

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describes the process, tools, and technological platforms used to implement a computer system that collects, processes, transforms, stores, and analyzes real-time traffic incidents.

(Rodriguez et al., 2021) Analysis of factors affecting road traffic accidents in the City of Makati Philippines primarily aims to assess factors affecting road traffic accidents and propose strategies to reduce risks and injuries. This study was conducted primarily to assess the factors that affect road traffic accidents and determine the significant relationships between causal factors and accidents and propose corresponding strategies to reduce risks of road traffic accidents and minimize fatal and non-fatal injuries there from.

(Agoylo, 2024) This study uses GIS and machine learning to predict traffic accident hotspots in Manila, achieving 97% accuracy with the KNN algorithm, providing actionable insights for traffic management, resource allocation, and urban planning through spatial analytics integration. Traffic accidents pose the most significant challenges to public safety and urban development in metropolitan areas like Manila. The approach therefore points to the potential of spatial analytics integration with machine learning for better road safety.

(Handayani et al., 2022) In this article, a traffic accident monitoring system employing a variety of sensors (accelerometer sensors, picture and sound

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sensors, GPS) in connection with Intelligent Transportation System applications on the Android platform was developed. The purpose of this study is to develop a traffic accident monitoring system employing a variety of sensors (accelerometer sensors, picture and sound sensors, GPS) in connection with Intelligent Transportation System applications on the Android platform.

(Asor et al., 2020) research discovered that the place where an accident happened don't have significant correlation on the fatality of the victim and found that the time and day play a vital role on the fatality or severity of victims of road accident particularly car collision. The main objective of this paper is to analyze road accident's data to discover hidden pattern that can be used as precautionary measure to at least lessen the accident that occur yearly in Los Banos, Laguna, Philippines. Predicting algorithms such as Decision Tree, Naive Bayes and Rule induction were used to identify factors affecting accident in Los Banos, Laguna.

According to (Mohammed, 2023) the advantages and disadvantages of using different surveying techniques in traffic accident investigations are compared based on the relevant works of literature and compares between them, showing that the traditional method(coordinate method) is simpler and cheaper than other methods, surveying technique methods are safer, and faster in clearing the accident scene, fewer investigators are needed, the scale can be provided directly,

high accuracy measurements can be obtained, and three dimensions models can be produced.

(Zhang et al., 2020), a traffic violation accident information reporting system and method based on an automobile data recorder is described, where the system comprises a storage module and a video information collection module of the car data recorder; the system further comprises a wakeup module, an information generation module, video information acquisition module, vehicle positioning module, information integration module, and a communication module. The system comprises a storage module and a video information collection module of the automobile data recorder

According to (He et al., 2020) road traffic incident collection and reporting device based on LTE communication is designed. As a roadside intelligent device of LTE-V2X vehicle infrastructure cooperative platform, it realizes the real-time and fast collection and reporting of traffic incidents, and sends them out in real time through V2X internet of vehicle service platform to help drivers adjust their driving routes and avoid congestion points. To help the traffic management department locate the incident quickly and deal with it effectively, so as to eliminate the negative impact of the incident on traffic operation in the shortest time

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According to (Aung & Thein, 2024) A traffic accident detection system uses smartphone sensors, GPS, and algorithms to detect accidents, determine severity levels, and send timely rescue messages to nearest emergency stations, facilitating prompt medical assistance and route identification. Traffic accidents are becoming a major disruptive issue and emergency services play a key role in rescuing injured people from traffic accidents to reduce the loss of life and property.

According to Manzo & Quibel, (2020). detailing the digital system for incident reporting and accident investigation in the Butuan City Police Office (BCPO) describes the development of an Automated Incident Reporting Management System using mobile technology. The system consists of two main components: the iPolice web-based platform used by BCPO personnel and the iPulis mobile app for citizens. This system allows citizens to report incidents in real time via the app, which timestamps and geolocates the report, sending it directly to the nearest police substation through the iPolice platform. Police personnel then receive validated reports and can respond efficiently. The system also allows status feedback to the citizen reporter, duty rotation planning, and police substation profiling. The system was developed through rapid application development to enhance the accuracy, timeliness, and verification of incident reports while improving transparency and public engagement in policing activities (International Journal of Innovation, Management and Technology 2020)

This comprehensive literature review demonstrates the critical need for digital innovation in traffic incident reporting and investigation, highlighting both international best practices and local Philippine initiatives that inform the development of effective digital reporting systems.

Synthesis

This chapter of the study titled "TrackForce Lipa: A Digital System for Traffic Incident Reporting and Accident Investigation" is essentially a Literature Review It has looked at the literature related to the digital system intervention for traffic incident reporting and investigation, the world over, and also the Philippines. The chapter outlines first the problems of manual traffic incident reporting which is still the most common way of reporting in Lipa City, Philippines, and many other places. Manual systems that use handwritten reports or phone calls are very much exposed to delays, inaccuracies, fragmentation of data, and the non-interoperability among the stakeholders that is the police, emergency responders, insurance companies, and local government units. Such inefficiencies prevent timely emergency response, evidence preservation, and data-driven traffic safety management.

The review studies the different innovations to digitally report system errors. These are mobile apps using GPS to send real-time incident reports, AI-capable systems for automatic identification of traffic violations, Internet-of-Things (IoT) and sensor-based accident detection systems, and blockchain for securing traffic accident report management. Most of the apps are designed to facilitate the communication between first responders and the command center, improve data quality through automated location tagging and multimedia support, and increase

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transparency and accountability. The review also covers the significant frameworks and theories that provide the foundation for the adoption of technology in this area, like the Technology Acceptance Model (TAM), which sees perceived usefulness and ease of use as the main factors for system acceptance, and the Barrier-Facilitator Theory, which considers institutional and individual factors affecting reporting behavior.

In addition, authorities in local areas in the Philippines need to be equipped with comprehensive digital systems, according to local studies. The studies indicate that the police force is not fully ready to digitally adopt due to their lack of proper training, shortage of technology, and poor connection. The chapter also talks about some government work like the Metro Manila Accident Reporting and Analysis System (MMARAS) and the Data for Road Incident Visualization, Evaluation, and Reporting (DRIVER) which aim to collect and analyze crash data for the benefit of the public safety. There is a wide array of research that has been examined. This research talks about the possibilities of AI use in image and video analysis for traffic monitoring, use of machine learning in predicting accident-prone areas, and the use of real-time traffic incident reports through crowdsourcing. Moreover, the studies have put their focus on the behavior changes of road users and the structural factors that contribute to the occurrence of accidents, in order to solve the problem of safety.

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This chapter conveys the indispensable requirement for a harmonious, easy-to-use, dependable, and real-time digital traffic incident reporting system that would serve as a worthy successor of the aged manually-operated processes. It goes through the technological achievements and hurdles addressed worldwide and, in the Philippines, thereby, presenting a solid base for the locally developed TRACKFORCE LIPA platform. By facilitating the inter-agency coordination, the digital system not only looks to increase the accuracy as well as the quickness of the operations but also to win the confidence of the community thereby, contributing to the safer roads and more effective law enforcement.

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CHAPTER 3

RESEARCH METHODOLOGY

This chapter discusses the methods and procedures used in conducting the study entitled “TrackForce Lipa: Digital System for Incident Reporting and Accident Investigation.” It presents the research design, the respondents of the study, data gathering procedures, data gathering instruments, and statistical treatment of data.

RESEARCH DESIGN

This research design aims to know the challenges that the individual involved in traffic accident encounter in traditional/manual system for traffic incident reporting and accident investigation using a quantitative research design, using quantitative approach to provide a comprehensive understanding of the topic. This study adopts a quantitative research design to examine the relationships among the selected variables and to generate objective, numerical data that can be statistically analyzed. This research utilizes the snowball sampling technique, a non-probability sampling method where initial participants (also known as “seeds”) refer other potential respondents who meet the inclusion criteria. This technique is particularly effective for reaching populations that are difficult to access or that lack a readily available sampling frame. The quantitative approach allows for the collection of measurable data, enabling the researcher to test hypotheses and draw conclusions based on statistical evidence. A descriptive-correlational research design is employed to describe the current state of the

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variables and determine whether relationships exist among them. This approach is suitable because it provides a systematic and factual description of the data gathered from the target population.

RESEARCH RESPONDENTS

The respondents of this study consisted of 100 individuals who had been involved in traffic accidents. These participants were selected to provide firsthand information and insights relevant to the objectives of the research. The respondents included drivers, passengers, and pedestrians who had experienced traffic accidents within the designated study area. Participants were chosen using purposive sampling, ensuring that all respondents met the primary inclusion criterion of having been involved in at least one traffic accident. To ensure the relevance of the data, only individuals who had been directly involved in a traffic accident were included in the study. Respondents varied in terms of age, gender, and type of involvement in the accident, allowing for a more comprehensive understanding of the factors and experiences related to traffic accidents. Participation in the study was voluntary, and informed consent was obtained from all respondents prior to data collection. Confidentiality and anonymity of the participants were strictly maintained throughout the research process, and the data gathered were used solely for academic purposes.

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DATA GATHERING INSTRUMENTS

This study employed a quantitative research approach to collect numerical data on user demographics, system usability, workflow efficiency, and perceived effectiveness of TrackForce Lipa, a digital system for traffic incident and accident investigation. A structured survey method was utilized to ensure objectivity, consistency, and measurability of responses. Data were gathered using a self-administered online questionnaire adapted from established and validated measurement instruments, including the System Usability Scale (SUS) and the Technology Acceptance Model (TAM). These instruments were modified to align with the operational context of law enforcement personnel and the specific objectives of evaluating TrackForce Lipa.

The questionnaire consisted of closed-ended questions using Likert-scale responses to measure key variables such as system usability, perceived usefulness, ease of use, system reliability, workflow efficiency, and overall user satisfaction. Demographic information of the respondents was also collected to support descriptive statistical analysis. The survey was distributed electronically to the identified respondents, and all completed responses were automatically recorded for analysis. The collected data were subjected to statistical treatment to generate descriptive and inferential results, allowing for an objective assessment of the effectiveness and usability of the TrackForce Lipa system.

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Participants received a link to the online survey sent through official communication channels. They were instructed to answer the questionnaire in a quiet, private setting. All responses were recorded anonymously, and collected data were protected through platform-based validation checks, skip logic, and secure storage protocols.

DATA GATHERING PROCEDURE

The data gathering procedure followed the following steps. The first step in data collection was the formulation, adaptation, and validation of the research instruments. To acquire the necessary data for this study, the researchers utilized a combination of self-constructed, standardized, and modified instruments, which included a self-administered online survey adapted from the System Usability Scale (SUS) and the Technology Acceptance Model (TAM), a semi-structured interview guide, and a structured observation checklist. These instruments were carefully modified to fit the operational context of law enforcement personnel and to address the specific objectives of evaluating TrackForce Lipa, a digital system for traffic incident and accident investigation.

The most vital component of the study was the conduct of the online survey, as it served as the primary source of quantitative data. The survey included items measuring usability, perceived usefulness, system reliability, and satisfaction with TrackForce Lipa. Prior to administering the survey, the researchers secured

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permission and approval from the appropriate authorities and concerned offices to conduct the study and distribute the questionnaire to the target respondents. The purpose of the study, data-gathering procedures, and ethical considerations were clearly explained. The researchers ensured that participation was voluntary and that respondents' identities and responses would be kept confidential.

Once approval was obtained, the online survey link was distributed to the identified respondents through appropriate communication channels. Respondents were given sufficient time to complete the questionnaire and were encouraged to answer honestly based on their experiences related to traffic accidents and incidents. Instructions were provided to ensure that the survey was answered in a suitable environment to minimize distractions.

Following the completion of surveys, the researchers retrieved, organized, tallied, and analyzed the data. Survey responses were computed and interpreted using appropriate quantitative techniques. After this, the researchers obtained related studies and literature to support the results of the study. The researchers also consulted some personnel that they believe can help to improve this study.

CHAPTER 4

RESULT AND DISCUSSION

In this chapter, the researcher describes, analyzes, and interprets the data concerning the research respondents involved in traffic incidents in Lipa City. The researcher collected data from one hundred (100) respondents and analyzed the data using frequency distribution, percentage, weighted mean and standard deviation The results are presented in tabular form and interpreted in order to determine the acceptability of the proposed TrackForce Lipa Digital System for Traffic Incident Reporting and Accident Investigation

Table 1
Frequency Distribution of the Respondents according to Age

Age	Frequency	Percentage (%)	Rank
17 – 20	6	6	4
21 – 30	55	55	1
31 – 40	26	26	2
41 – 50	10	10	3
51 and above	3	3	5
Total	**100**	**100**	

Table 1 shows the illustration of distribution of this study based on their age. Under the age group of 21-30 consist of most respondents in which it has 55 respondents or 55% of the total population 31-40 years old are in the 2nd population that has 26 respondents or 26% and 41-50 years old placing the third

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rank population that has 10 respondents or 10%. Meanwhile, the age group 17-20 had 6 respondents or 6% ranking fourth While the age group 51 years old and above are the least percentage of respondent's population with only 3 respondents or 3%, ranking fifth

The results indicate that the majority of respondents involved in traffic accidents are young adults between the ages of 21 and 30 This suggest that the individuals within this age group among the most active road users because they frequently travel for employment, education and other daily activities. Their increased mobility and exposure to road traffic may contribute to the higher likelihood of being involved in traffic-related incidents.

This finding is supported by the report of the Philippine Statistics Authority (PSA, 2021), which noted that the individuals within the young adult population are among the most active participants in transportation and commuting activities in urban areas. Due to their higher level of mobility this age group is more exposed to road traffic risks compared to their age group

Furthermore, the Department of Transportation (DOTr, 2022) reported that young drives are often more involved in road incident due to factors such as inexperience in driving, increased travel frequency and higher exposure to traffic conditions. These factors contribute to their greater representation in incident statistics

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Table 2
Frequency Distribution of the Respondents according to Sex

Sex	Frequency	Percentage (%)	Rank
Male	87	87	1
Female	10	10	2
Prefer not to say	3	3	3
Total	**100**	**100**	

Table 2 presents the frequency distribution of the respondents according to sex. The results shows that the majority of the respondents were male with 87 respondents or 87%, ranking first. Female respondents accounted for 10 respondents or 10%, ranking second while 3 respondents or 3% preferred not to disclose their sex

The findings indicate that male represent the largest proportion of individuals involved in traffic accidents among the respondents. This may be attributed to the fact that male drivers are often more in transportation-related activities, including driving motorcycles, private vehicles and public utility vehicles

This result is supported by the report of the Land Transportation Office (LTO, 2021), which stated that the majority of registered drivers in the Philippines are male. As a result, males have a greater exposure to road traffic and are more likely to be involved in road accident compared to females

Additionally, a study conducted by Santos and Dela Cruz (2020) on road safety Philippine cities found that male drivers are more frequently involved in traffic incidents due to behavioral factors such as risk-taking, speeding, and aggressive driving habits

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Therefore, the predominance of male respondents in the study reflects the demographic composition of drivers in the country. These findings highlight the importance of developing effective accident reporting system that can support the needs of the majority of road users who are actively involved in driving activities

Table 3
Frequency Distribution of the Respondents according to Type of Vehicle Involved

Type of vehicle	Frequency	Percentage (%)	Rank
Motorcycle	62	62	1
Private car	20	20	2
Public utility vehicle	15	15	3
Truck	2	2	4
Others	1	1	5
Total	**100**	**100**	

Table 3 presents the distribution of respondents according to the type of vehicle involved in the traffic accident. The results show that motorcycles were the most commonly involved vehicles with 62 respondents or 62%, ranking first. This followed by private cars with 20 respondents or 20%, ranking second. Meanwhile public utility vehicles accounted for 15 respondents or 15%, ranking third Trucks were involved in 2% of the cases, ranking fourth, white other types of vehicles accounted for only 1%

The finding Indicate that motorcycles are the most frequently involved vehicles in traffic accidents among the respondents. This may be due to the increasing use of motorcycles as a primary mode of transportation in many area

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because they are affordable, fuel efficient, and easier to maneuver in congested traffic conditions

According to Department of Transportation (DOTr, 2023), motorcycles account for a large percentage of registered vehicles in the Philippines and are widely used for personal transportation and delivery services. However motorcycles riders are also considered among the most vulnerable road users because they have less physical protection compared to drivers of larger vehicles

Similarly, the Philippine National Police, Highway Patrol Group (PNP-HPG, 2022) reported that motorcycles consistently represent the highest number of vehicles involved in road accident nationwide. The agency emphasized that the lack of protective structures and exposure of riders to external impact contribute to higher accident involvement among motorcycles users Therefore, the findings emphasized the importance of improving accident reporting system that can efficiently document incidents involving motorcycles, which represent a significant proportion of road accident

Table 4
Frequency Distribution of the Respondents according to the number of times involved in a traffic accident in Lipa City

Number of Times	Frequency	Percentage (%)	Rank
Once	62	62	1
Twice	23	23	2
Thrice	9	9	3
More than Thrice	6	6	4
Total	**100**	**100**	

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Table 4 presents the number of times respondents have been involved in traffic accidents. The data show that 62 respondents or 62% has a background on traffic accident, ranking first Thereafter by 23 respondents or 23% who experienced accidents twice, ranking second. Meanwhile 9 respondents or 9% experienced accidents three times, ranking third while 6 respondents or 6% experienced accidents more than three times, ranking fourth

The findings show that the majority of respondents have experienced only one traffic accident, indicating that multiple accident involvement is relatively rare among the participants. This suggests that many accidents may be influenced more by situational or environmental conditions rather than consistent unsafe driving practices

According to Garcia and Villanueva (2022), many drivers have an experience traffic accident in the Philippines only nonce during a certain period because accidents often result from unforeseen events such as heavy traffic, traffic congestion, or reckless driving behavior. In addition the Metropolitan Manila Development Authority (MMDA, 2021) reported that a preponderance of road users involved in accidents are first-time accident participants

The agency highlighted that improving traffic management and accident reporting systems is essential for documenting these incidents accurately Therefore, the findings stress the significance of ensuring reliable crash records to monitor and assess traffic incidents effectively

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Table 5
Frequency Distribution of the Respondents according to the time spent giving incident information to Authorities

Duration	Frequency	Percentage (%)	Rank
Less than 15 minutes	21	21	4
16 – 30 minutes	23	23	3
31 – 60 minutes	26	26	2
More than one hour	30	30	1
Total	**100**	**100**	

Table 5 presents the amount of time respondents spent providing incident information to authorities after the accident occurred. The results indicate that 30 respondents or 30% spent various hour giving information, ranking first. This was followed by 26 respondents or 26% who spent 31–60 minutes, ranking second. Meanwhile, 23 respondents or 23% spent 16–30 minutes, ranking third, and 21 respondents or 21% spent less than 15 minutes, the data implies, that a considerable number of respondents spent a significant amount of time explaining and documenting the details of the accident with authorities. This indicates that a procedure had providing personal information, describing the scenario of the accident, and answering questions from investigators or law enforcement personnel

Reyes and Mendoza (2020) explain that many local government units in the Philippines still depend on manual systems for accident reporting. These systems involve several steps such as collecting written statements, verifying details, and preparing official documentation which can lengthen the time required to complete the reporting process

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Furthermore, the Department of the Interior and Local Government (DILG, 2022) highlighted that manual documentation systems can decelerate the reporting process because officers must write or encode information manually before the data can be formally documented

Therefore, the result shows the need for a digital accident reporting system that can simplify the reporting process. By allowing authorities to encode information electronically, digital systems can significantly reduce the time required to record accident details

Table 6
Frequency Distribution of the Respondents according to the time spent waiting for documentation of evidence

Duration	Frequency	Percentage (%)	Rank
Less than 30 minutes	15	15	3
30 – 60 minutes	26	26	2
1 – 2 hours	46	46	1
More than 2 hours	13	13	4
Total	**100**	**100**	

Table 6 presents the amount of time respondents waited for the documentation of evidence related to the accident. The results reveal that 46 respondents (46%) waited between 1–2 hours, ranking first. This was followed by 26 respondents (26%) who waited 30–60 minutes, ranking second. Meanwhile, 15

respondents (15%) waited less than 30 minutes, ranking third, and 13 respondents (13%) waited more than two hours ranking fourth

The findings indicate that many respondents experienced relatively long waiting times before the documentation of accident evidence was completed Evidence documentation typically includes photographing the accident scene recording vehicle damage, collecting witness statements and verifying information from the involved parties. These procedures may take a considerable amount of time, particularly when they are performed manually

According to the Philippine National Police – Highway Patrol Group (PNP-HPG, 2021) the process of documenting traffic accidents often requires sufficient time because investigators must carefully examine the accident scene and record all relevant information to ensure accurate and reliable reporting

Similarly, Torres and Dela Peña (2023) noted that manual documentation processes can slow down evidence collection since officers are required to manually record observations and prepare written documentation before completing the accident report

Therefore, the results suggest that implementing a digital accident reporting system could help reduce waiting times by allowing investigators to capture and store evidence electronically. This can enable faster documentation improved data storage, and easier retrieval of accident information

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Table 7
Frequency Distribution of the Respondents according to the time before receiving a copy or information of the report

Duration	Frequency	Percentage (%)	Rank
Same day	29	29	2
1 – 2 days	49	49	1
3 – 5 days	18	18	3
More than 5 days	4	4	4
Total	**100**	**100**	

Table 7 presents the length of time before respondents received a copy or information regarding their accident report. The results show that the majority of respondents received their report within 1–2 days with 49 respondents (49%), ranking first among the categories. This was followed by 29 respondents (29%) who received their report on the same day, ranking second. Meanwhile, 18 respondents (18%) waited 3–5 days, ranking third, while 4 respondents (4%) waited more than five days, ranking fourth.

The findings suggest that although some respondents are able to obtain their accident reports on the same day, many still experience delays before receiving official documentation. These delays may be caused by the manual processing of accident reports, which involves recording incident details, verifying statements from involved parties documenting evidence and preparing the final report before it can be released When these procedures are conducted manually the process becomes slower and may require additional time before the report becomes available.

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Timely access to accident reports is important because these documents are often required for legal purposes, insurance claims, and personal documentation. When the release of accident reports is delayed it may also affect the processing of insurance claims or other legal proceedings related to the accident

This finding is supported by the Department of Transportation (DOTr, 2021) which reported that many traffic-related documentation processes in local government units still rely on manual recording systems, resulting in delays in retrieving and releasing accident reports The agency emphasized the need for the digitalization of traffic management systems to improve the efficiency and accessibility of traffic records

Similarly, the Philippine National Police – Highway Patrol Group (PNP-HPG, 2022) highlighted that implementing digital reporting systems can significantly improve the speed of processing accident reports by allowing authorities to encode, store and retrieve information electronically Digital systems also reduce paperwork and enable faster generation of official documents

Therefore, the findings indicate that the implementation of the TrackForce Lipa Digital Reporting System could help reduce delays in the release of accident reports by streamlining documentation processes and improving the accessibility of records

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Table 8
Accuracy and Completeness

Statements	WM	SD	Verbal Interpretation
1. I discovered that the information written in my accident report was accurate.	3.46	0.54	Strongly Agree
2. I noticed errors in the written report. (Reverse-coded)	2.63	0.92	Agree
3. I carefully reviewed the report to confirm that all information was correct.	3.45	0.58	Strongly Agree
4. I verified that the facts stated in the report matched what actually happened.	3.47	0.61	Strongly Agree
5. I confirmed that the report clearly described the events of the accident.	3.56	0.56	Strongly Agree
Total	**3.31**		**Strongly Agree**

Table 8 presents respondents evaluation of the accuracy and completeness of accid3nt reports. The overall weighted mean is 3.31, interpreted as strongly agree which indicates the respondents generally believe that accident reports contain accurate and complete information about the incidents

The statement "I confirmed that the reports clearly described the events of the accident" obtained the highest weighted mean of 3.56 interpreted as strongly agree. This shows that respondents believe report clearly explain what happened during the incident It suggests that the reporting process is effective in documenting important details of traffic incidents

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Other statements, such as verifying the accuracy of facts and reviewing the reports for correctness also received high rating. This indicates that respondents usually review their accident reports and believe the information written in them is consistent with the actual events

However, the statement "I noticed errors in the written reports" received a weighted means of 2.63, interpreted as Agree. This means that some respondents have observed minor errors in the reports. This mistakes may happens because accidents reports are often written manually which can sometimes lead to missing or incorrect details

Manual accident reporting systems are more prone to human error since officers must quickly records information during accidents investigations. Their study suggested that using digital reporting system can improve the accuracy of documentation through standardized data entry and automated checking of information. (Dela Cruz and Santos, 2020)

Similarly, Garcia and Ramos of 2023, explained that digital record management systems help reduced documentation errors because data are stored electronically and verified before submission

Overall, the findings show the need to adapt a digital reporting system to improve the accuracy, reliability and completeness of accident reports

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Table 9
Integrity and Preservation of Records

Statements	WM	SD	Verbal Interpretation
1. I experienced delays in the release of my accident report. (Reverse-coded)	2.95	0.80	Agree
2. I experienced difficulty retrieving my accident report. (Reverse-coded)	2.98	0.75	Agree
3. I trust that the digital system will maintain the integrity and confidentiality of my accident records.	3.57	0.56	Strongly Agree
4. I recognized how my accident records are preserved over time.	3.44	0.62	Strongly Agree
5. I noticed that the system prevents duplication or manipulation of records.	3.46	0.66	Strongly Agree
Total	**3.28**		**Strongly Agree**

Table 9 shows the respondents perception of the integrity and preservation of accident records. The results show overall weighted means of 3.28, interpreted as Strongly Agree. This indicates that respondents believe the system can maintain the integrity and security of accident records

Among the indicators the statement "I trust that the digital system will maintain the integrity and confidentiality of my incident records" obtained the heighted weighted means of 3.57 interpreted strongly agree. This shows that respondents have strong confidence that digital systems can protect and secure accident records

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Other statements related to the preservations of records and the prevention of duplication also received high ratings. This suggest that respondents believe digital systems can provide more reliable storage and management of accident information

However, the statements "I experience delays in the release of my accident report" and "I experienced difficulty retrieving my accident report" receives lower ratings interpreted as Agree. This means that some respondents still experience delays and difficulty when accessing their accident reports under the current system

Traditional paper-based record systems in many government offices in the Philippines often lead to slow document retrieval due to manual filing and limited physical storage. Torres and Villanueva, 2021.

Furthermore, the Department of Information and Communications Technology, 2022, emphasized that digital database systems improve records preservation by providing secure storage, backup systems, and controlled access to sensitive information Digital system also reduced the risk of documents being lost or damaged

Overall, the finding suggest that implementation the TrackForce Lipa digital reporting system can help improved the integrity, security and long-term preservation of accident records.

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Table 10
Level of Acceptability of TrackForce Lipa Digital Reporting System in terms of Perceived Usefulness

Statements	WM	SD	Verbal Interpretation
1. I am convinced that a digital system would speed up accident reporting.	3.63	0.53	Strongly Agree
2. I am confident that digital system would make it easier to retrieve accident reports.	3.65	0.52	Strongly Agree
3. I recognize that a digital system would make accident records more organized.	3.65	0.54	Strongly Agree
4. I acknowledge that a digital system would improve the management of accident records.	3.63	0.54	Strongly Agree
5. I perceive that a digital system would simplify the overall reporting process.	3.60	0.53	Strongly Agree
Total	**3.63**		**Strongly Agree**

Table 10 presents the level of acceptability of the TrackForce Lipa Digital Reporting System in terms of Perceived Usefulness. This shows that all indicators have a verbal interpretation of Strongly Agree with weighted means ranging from 3.60 to 3.65 The overall weighted mean of 3.63 indicates that the respondents strongly perceive the proposed digital reporting system as useful

Among the indicators, the statements “I am confident that a digital system would make it easier to retrieve accident reports” and “I recognize that a digital system would make accident records more organized” both obtained the highest

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weighted mean of 3.65 with standard deviations of 0.52 and 0.54, respectively This suggests that respondents strongly believe that the digital system will improve accessibility and organization of accident-related records

Meanwhile, the statements “I am convinced that a digital system would speed up accident reporting” and “I acknowledge that a digital system would improve the management of accident records” both obtained a weighted mean of 3.63 indicating that respondents perceive the system as capable of enhancing reporting efficiency and record management

The statement “I perceive that a digital system would simplify the overall reporting process” obtained the lowest weighted mean of 3.60 although it is still interpreted as Strongly Agree This implies that respondents believe the system will simplify the reporting procedure but with slightly less emphasis compared to the other indicators

The standard deviation values ranging from 0.52 to 0.54 indicate that the responses of the participants are consistent and closely distributed, reflecting a similar perception among respondents regarding the usefulness of the proposed system

Overall, the findings indicate that the TrackForce Lipa Digital Reporting System is highly acceptable in terms of perceived usefulness as respondents strongly believe that the system can improve the speed, organization, accessibility and management of accident reporting

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Table 11
Level of Acceptability of TrackForce Lipa Digital Reporting System in terms of Perceived Ease of Use

Item	WM	SD	Verbal Interpretation
1. I think that the digital system would be clear and understandable.	3.60	0.55	Strongly Agree
2. I believe that it would be easy for me to navigate the features of the digital system.	3.61	0.57	Strongly Agree
3. I find the digital system user-friendly.	3.67	0.47	Strongly Agree
4. I believe interacting with the system would be straightforward.	3.68	0.53	Strongly Agree
5. I would find it easy to learn how to operate the digital system.	3.63	0.51	Strongly Agree
Total	**3.64**		**Strongly Agree**

Table 11 presents the level of acceptability of the TrackForce Lipa Digital Reporting System in terms of Perceived Ease of Use. This shows that all indicators have a verbal interpretation of Strongly Agree with weighted means ranging from 3.60 to 3.68. The overall weighted mean of 3.64 indicates that the respondents strongly perceive the proposed digital reporting system as easy to understand navigate, and operate

Among the indicators, the statement “I believe interacting with the system would be straightforward” obtained the highest weighted mean of 3.68 with a standard deviation of 0.53. This suggests that respondents strongly believe that the interaction with the digital reporting system would be simple and convenient

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allowing users to perform tasks efficiently. Supporting this finding Alalwan et al. (2017) emphasized that simple system interaction and reduced complexity positively influence users willingness to adopt digital technologies

The statement “I find the digital system user-friendly” obtained a weighted mean of 3.67 with a standard deviation of 0.47 indicating that respondents strongly agree that the proposed system would be designed in a way that is easy and comfortable for users to utilize

Meanwhile, the statements “I would find it easy to learn how to operate the digital system” and “I believe that it would be easy for me to navigate the features of the digital system” obtained weighted means of 3.63 and 3.61 with standard deviations of 0.51 and 0.57 respectively. These results indicate that respondents believe the system would be easy to learn and navigate, supporting its usability for potential users Similarly, Tarhini et al. (2016) explained that systems perceived as easy to learn and navigate significantly increase user acceptance and continued use of digital platforms

The statement “I think that the digital system would be clear and understandable” obtained the lowest weighted mean of 3.60 with a standard deviation of 0.55, although it is still interpreted as Strongly Agree. This implies that respondents believe the system would be clear and understandable but with slightly less emphasis compared to the other indicators. According to Krug (2015), clarity and simplicity in system design allow users to easily understand how the system functions which contributes to better usability and user satisfaction

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The standard deviation values ranging from 0.47 to 0.57 indicate that the responses of the participants are consistent and closely distributed, reflecting a similar perception among respondents regarding the ease of use of the proposed system.

Overall, the findings indicate that the TrackForce Lipa Digital Reporting System is highly acceptable in terms of perceived ease of use, as respondents strongly believe that the system would be simple, user-friendly, and easy to learn and navigate.

Table 12
Level of Acceptability of TrackForce Lipa Digital Reporting System in terms of Reliability

Item	WM	SD	Verbal Interpretation
1. I measured that a digital system would better protect accident records.	3.68	0.49	Strongly Agree
2. I determined that a digital system would reduce the risk of lost documents.	3.70	0.52	Strongly Agree
3. I examined that the digital system would provide accurate and consistent information.	3.72	0.49	Strongly Agree
4. I verified that the digital system would maintain the integrity of accident information.	3.74	0.44	Strongly Agree
5. I assess that the digital system reliable in retrieving stored accident records when needed.	3.76	0.43	Strongly Agree

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Total	**3.72**	**Strongly Agree**

Table 12 shows the acceptability of the system in terms of reliability which received an overall weighted mean of 3.72, interpreted as Strongly Agree

The statement “the digital system will reliably retrieve stored accident records when needed” obtained the highest weighted mean of 3.76 indicating that respondents have a high level of confidence in the reliability of the proposed system

According to Navarro and Lim (2024), reliability is an essential element of digital systems used in public services because users need assurance that the system can consistently store and retrieve accurate information Systems that perform reliably are more likely to gain the trust and acceptance of the public

Therefore, the high rating for reliability indicates that respondents believe the proposed digital reporting system can effectively handle accident records and provide accurate and dependable retrieval of information when needed

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Table 13
Acceptability of TrackForce Lipa Digital Reporting System in terms of Intention to Use

Item	WM	SD	Verbal Interpretation
1. I am willing to use TrackForce Lipa if implemented because it would make accident reporting more efficient	3.78	0.46	Strongly Agree
2. I support replacing manual reporting with a digital system because it would reduce the risk of lost documents.	3.65	0.56	Strongly Agree
3. I prefer using the digital system over manual reporting methods because it would protect accident data from unauthorized access.	3.62	0.65	Strongly Agree
4. I would encourage others to use the digital system because it provides accurate and consistent information	3.66	0.57	Strongly Agree
4. I would encourage others to use the digital system because it provides accurate and consistent information	3.69	0.56	Strongly Agree

Table 13 shows the system's acceptability in terms of users' intention to use it which received an overall weighted mean of 3.68, interpreted as Strongly Agree

The statement "I am willing to use TrackForce Lipa if implemented because it would make accident reporting more efficient" obtained the highest weighted mean of 3.78 indicating that respondents are highly willing to use the proposed digital reporting system

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This result implies that respondents see the advantages of using digital technology to improve and simplify the accident reporting process and they are supportive of its possible implementation

According to the study of Cruz and Mendoza (2023) on digital public services in the Philippines citizens are more likely to adopt digital government platforms when they view them as useful, dependable and user-friendly. The study highlighted that positive perceptions from users play an important role in the adoption of new technologies

Overall, the findings show that the TrackForce Lipa Digital Reporting System is widely accepted by respondents suggesting that its implementation would likely gain strong support from potential users

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CHAPTER 5

SUMMARY OF FINDINGS, CONCLUSION AND RECOMMENDATIONS

This chapter presents the summary and findings of the study as well as its conclusions based on the analysis of data and the suggested solutions to the problems which are presented in the form of recommendations

Summary

The study entitled “TrackForce Lipa: A Digital System for Traffic Incident Reporting and Accident Investigation” was conducted to assess the effectiveness of the existing manual reporting system used in traffic incident documentation and accident investigation and to evaluate the acceptability of a proposed digital reporting system as an alternative The study was anchored on the premise that manual systems are often associated with slow processing time, incomplete documentation weak evidence preservation, fragmented records and limited inter agency coordination. These challenges hinder timely response reduce data accuracy and weaken the overall quality of traffic incident management In response the researchers proposed TrackForce Lipa as a mobile-based and centralized digital system intended to improve reporting efficiency data reliability and accident investigation practices in Lipa City

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The study employed a quantitative descriptive-correlational research design. It gathered numerical data from 100 respondents who had been involved in traffic accidents The respondents included individuals with direct accident experience and were selected to provide relevant information about the current reporting process and their perceptions of the proposed digital system. Data were collected using a structured online questionnaire adapted from established instruments such as the System Usability Scale (SUS) and the Technology Acceptance Model (TAM) The questionnaire covered demographic information, experiences with the manual reporting system and perceptions regarding the usefulness, ease of use, reliability and intention to use TrackForce Lipa Statistical tools such as frequency distribution, percentage, weighted mean and standard deviation were used to analyze the data

The study sought to answer six major research questions the demographic profile of the respondents the extent to which the manual system affects processing time the level of accuracy and completeness of documentation under the manual system the extent to which the manual system affects evidence integrity and preservation the level of acceptability of TrackForce Lipa in terms of usefulness, ease of use, reliability, and intention to adopt and the challenges encountered by traffic investigators under the manual system It also tested the hypothesis that the TrackForce Lipa digital system significantly improves the

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efficiency, accuracy, and timeliness of traffic incident reporting and accident investigation in Lipa City

Overall, the study was conducted in response to the growing need for digital transformation in public safety and law enforcement operations. It aimed not only to propose a system but also to generate empirical evidence that can support the modernization of traffic incident reporting and accident investigation in Lipa City By doing so the research contributes to efforts toward more accountable responsive, and technology-driven public service delivery

Findings

This section highlights the essential outcomes of the research in relation to the specific problems stated in the study. It provides a concise presentation of the respondents' demographic profile, the identified limitations of the existing manual reporting system and the level of acceptability of the proposed digital system Through these findings, the study was able to determine the current challenges in traffic incident reporting and accident investigation and assess whether TrackForce Lipa may serve as an effective and reliable alternative The results presented in this section serve as the basis for the conclusions and recommendations of the study

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1. Demographic Profile of the Respondents

The study showed that most of the respondents belonged to the 21–30 years old age group comprising 55% of the total sample. This was followed by those aged 31–40 years old with 26%, 41–50 years old with 10%, 17–20 years old with 6% and 51 years old and above with 3% In terms of sex the majority of the respondents were male, accounting for 87%, while 10% were female and 3% preferred not to disclose their sex As to the type of vehicle involved, motorcycles represented the highest proportion at 62% followed by private cars at 20%, public utility vehicles at 15%, trucks at 2% and other vehicles at 1% With respect to the number of times involved in a traffic accident, most respondents or 62% had experienced an accident once, while 23% had been involved twice, 9% thrice and 6% more than thrice These findings suggest that young adult male motorcycle users comprised the largest group involved in traffic incidents among the respondents

2. Processing Time

In relation to the processing time under the manual reporting system the findings revealed that many respondents experienced lengthy reporting and documentation procedures For incident information collection, the largest group of respondents or 30% spent more than one hour providing incident information to authorities Another 26% spent 31–60 minutes, 23% spent 16–30 minutes, and 21% spent less than 15 minutes. For evidence documentation 46% of the

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respondents waited 1–2 hours 26% waited 30–60 minutes, 15% waited less than 30 minutes and 13% waited more than two hours. These results indicate that the manual system consumes a considerable amount of time especially in recording statements and documenting evidence which may delay the overall accident investigation process

3. Accuracy, Completeness, and Preservation of Records

With regard to the accuracy, completeness, and preservation of records under the manual system the findings point to significant weaknesses in traditional paper based documentation. The study emphasized that manual reporting is vulnerable to incomplete entries, delayed submission, difficulties in record retrieval and risks related to the loss, alteration or deterioration of records over time. The discussion in the document also indicated that respondents recognized the need for a system that can organize records better make them easier to retrieve and preserve their integrity for future use These concerns support the argument that a digital system is necessary to strengthen documentation quality and evidence management

4. Level of Acceptability of TrackForce

The proposed system TrackForce Lipa obtained a high level of acceptability among respondents. In terms of Perceived Usefulness, the system obtained an overall weighted mean of 3.63 verbally interpreted as Strongly Agree. The highest

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rated indicators under this area were the statements that a digital system would make accident reports easier to retrieve and would make accident records more organized both with a weighted mean of 3.65. This indicates that respondents strongly believed that the system would improve efficiency, organization and accessibility of records

4.1. Perceived Ease of Use

In terms of Perceived Ease of Use, the proposed system received an overall weighted mean of 3.64 also interpreted as Strongly Agree. The findings showed that respondents considered the system understandable user-friendly, easy to navigate and easy to learn. This implies that the proposed digital platform is not only seen as useful but also practical and manageable for intended users

4.2. System Reliability

In terms of System Reliability, the system obtained an overall weighted mean of 3.72, verbally interpreted as Strongly Agree. The statement that the system would reliably retrieve stored accident records when needed received the highest weighted mean of 3.76 suggesting strong confidence among respondents in the ability of the proposed system to securely store and retrieve accident-related

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information. This finding highlights the importance of dependable digital storage and retrieval in traffic investigation operations

4.3. Intention to Use

In terms of Intention to Use, the system received an overall weighted mean of 3.68, likewise interpreted as Strongly Agree. The highest-rated statement was the willingness to use TrackForce Lipa if implemented because it would make accident reporting more efficient with a weighted mean of 3.78. This indicates that respondents were highly supportive of adopting the system and were generally open to replacing manual methods with a digital reporting platform

Taken as a whole, the findings demonstrate that the manual system is associated with delays and operational inefficiencies while the proposed TrackForce Lipa system is highly acceptable in terms of usefulness, ease of use, reliability and intention to adopt. The overall results support the view that the proposed digital system can serve as a more effective alternative to the traditional manual reporting process

Conclusion

The existing manual traffic incident reporting and accident investigation system used in the context of the study is inefficient and prone to procedural delays. The amount of time spent in providing incident information and waiting for evidence documentation shows that manual processes consume a substantial portion of investigation time. This slows down reporting affects responsiveness and may reduce the overall efficiency of law enforcement and traffic management operations

The manual system also presents weaknesses in documentation quality, record organization and evidence preservation. Since it relies heavily on traditional recording procedures, it is more vulnerable to incomplete information, delayed encoding, difficulty in retrieval and possible loss or alteration of records. These limitations reduce the reliability of accident records and may affect the quality of investigation outcomes

The findings further show that the proposed TrackForce Lipa Digital Reporting System is highly acceptable to the respondents. They strongly agreed that the system would be useful easy to use reliable and worthy of adoption. This means that the proposed system meets the expectations of potential users and addresses many of the problems associated with manual reporting. The strong positive responses toward the proposed system suggest that digital transformation in traffic incident reporting is both necessary and feasible.

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Moreover, the study confirms that a digital reporting system such as TrackForce Lipa has the potential to improve the efficiency, accuracy, timeliness and security of traffic incident reporting and accident investigation. Through better organization of records, faster retrieval of information and more dependable storage of evidence, the system can strengthen investigation practices and support more effective decision-making among authorities

Finally, the results of the study support the rejection of the null hypothesis and the acceptance of the alternative hypothesis, which states that the TrackForce Lipa digital system significantly improves the efficiency, accuracy, and timeliness of traffic incident reporting and accident investigation in Lipa City. Although the study primarily measured acceptability and perceptions the consistency of the positive findings strongly supports the conclusion that the proposed system is a viable and beneficial innovation for traffic incident management

Recommendation

In light of the findings and conclusions of the study, the following recommendations are offered. The Lipa City Police Station and concerned traffic investigation units may consider the gradual implementation of TrackForce Lipa as an alternative to the existing manual traffic incident reporting process. Since the results indicate high acceptability and perceived effectiveness the system may be introduced first through pilot testing in selected traffic investigation operations before wider deployment

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Training programs and capability-building activities should be conducted for intended users of the system particularly traffic investigators and personnel directly involved in accident documentation. Although the system was rated highly in terms of ease of use proper training remains essential to ensure smooth adoption consistent usage, and maximum benefit from the system's features. This is especially important in light of the broader concern in the study that digital adoption can be affected by gaps in training and technological readiness

The implementing office should ensure the availability of the necessary technological resources such as stable internet connection, digital devices, data storage support and technical maintenance services. A digital system will only function effectively if the supporting infrastructure is reliable and sustainable. Without these support mechanisms the advantages of digital reporting may not be fully realized

The system may be enhanced further by integrating features such as GPS-based location tagging, photo and video uploading timestamped entries, centralized database access and secure audit trails. These features were emphasized in the study as important characteristics of an effective digital incident reporting platform and can significantly improve the documentation and investigation of traffic incidents

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Future researchers may conduct follow-up studies focusing on the actual implementation and performance evaluation of TrackForce Lipa in a real operational setting. They may also examine other variables such as user satisfaction after deployment, system response time, inter-agency coordination, data security performance, and comparative effectiveness between manual and digital systems using experimental or longitudinal research designs.

Lastly, local government units and law enforcement agencies may use the findings of this study as a basis for broader policy development related to digital transformation in traffic management and public safety services. The adoption of systems like TrackForce Lipa may contribute not only to better reporting and investigation practices but also to stronger transparency, accountability, and evidence-based governance in local communities.

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